

Unit 13, Lesson 2: Circles on the Coordinate Plane



Benchmarks

MA.912.GR.3.2 Given a mathematical context, use coordinate geometry to classify or justify definitions, properties, and theorems involving circles, triangles, or quadrilaterals.

MA.912.GR.3.3 Use coordinate geometry to solve mathematical and real-world geometric problems involving lines, circles, triangles, and quadrilaterals.

MA.912.GR.3.4 Use coordinate geometry to solve mathematical and real-world problems on the coordinate plane involving perimeter or area of polygons.



Learning Targets

- ☐ I can use coordinate geometry to determine parts of a circle in a mathematical context.
- ☐ I can draw a circle using given definitions, properties, or theorems in a mathematical context.
- ☐ I can use coordinate geometry to solve mathematical problems involving perimeter and area of polygons.



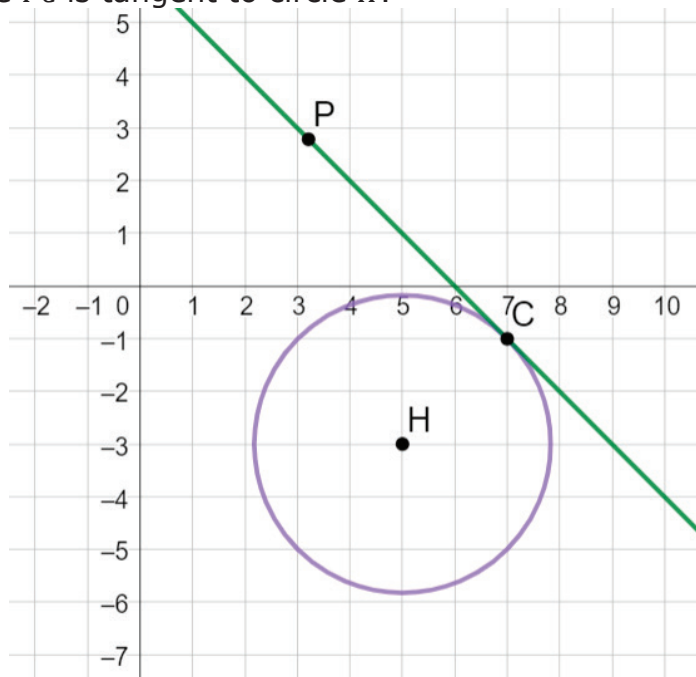
13.2.1 Warm-Up: Bell Work

1. Which equation represents a line that is perpendicular to the line $y = -\frac{1}{3}x + 3$?
 - A. $3x + y = 4$
 - B. $x - 3y = 24$
 - C. $y - 3x = -4$
 - D. $x + 3y = 3$



13.2.2 Exploration: Circles on the Coordinate Plane

1. Circle H is shown, where point C is on the circle and line PC is tangent to circle H .

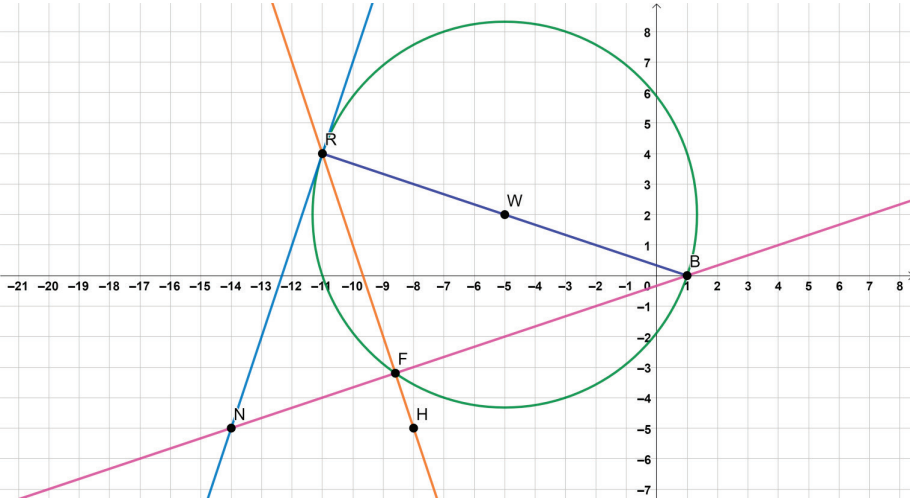


- Use a straightedge to draw a line that goes through points C and H .
- Discuss with your partner how the slope of tangent line PC and the slope of the radius CH are related.
- What property of circles explains this relationship?

- Write the equation of line PC .
- Write the equation of line CH .
- Line PC and line CH intersect at point $C (7, -1)$. Use the equations of lines PC and CH to confirm this point.

2. Circle M has a center at $(-2, 1)$ and a tangent line with the equation $y = 2x - 5$.
- Write the equation of the line perpendicular to the tangent line.
 - Find the point where the tangent line touches circle M . Call this point R .
 - What is the radius of circle M ?
 - Write the equation of circle M .

3. Circle W is shown, with secant lines RH and NB , tangent line NR , and chord RB .



a. Complete the table.

Line Segment	Length
$\overline{NB}$	
$\overline{NR}$	

b. Determine the length of $\overline{NF}$.

c. Complete the table.

Line	Slope	Equation in Point-Slope Form
$\overleftrightarrow{NR}$		
$\overleftrightarrow{RH}$		
$\overleftrightarrow{NB}$		

d. Rewrite the equation of $\overleftrightarrow{RH}$ in slope-intercept form.

e. Rewrite the equation of $\overleftrightarrow{NB}$ in slope-intercept form.

f. Use the slope-intercept forms of $\overleftrightarrow{RH}$ and $\overleftrightarrow{NB}$ to find point F .



- g. Work with your partner to determine if $\triangle BRN$ is isosceles. Show your work as justification.

- h. What could be used to justify that $\triangle RFB$ is a right triangle?

- i. What is the area of $\triangle RFB$?

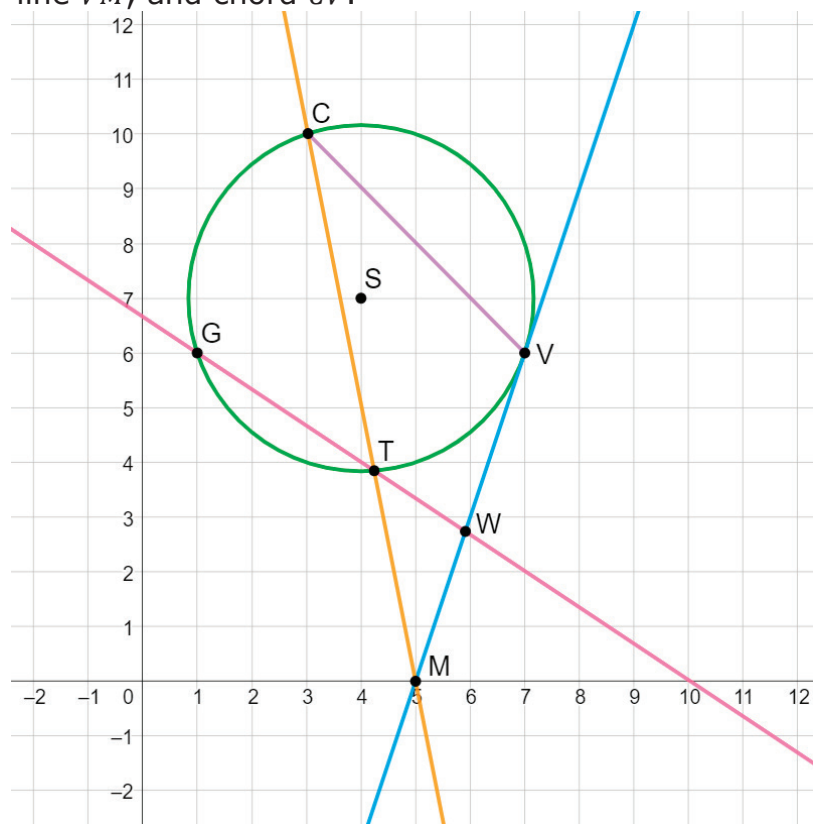
- j. What is the perimeter of $\triangle RFN$?



13.2.3 Try It: Circles on the Coordinate Plane

1. Circle G has a center at $(9, -2)$ and a tangent line with the equation $3x + 2y = 36$.
 - a. At what point does the tangent line touch circle G ?
 - b. Write the equation of circle G .

2. Circle S is shown with secant lines CM and GW , tangent line VM , and chord CV .



- a. Complete the table.

Line Segment	Length
$\overline{VM}$	
$\overline{CV}$	
$\overline{CM}$	
$\overline{MT}$	

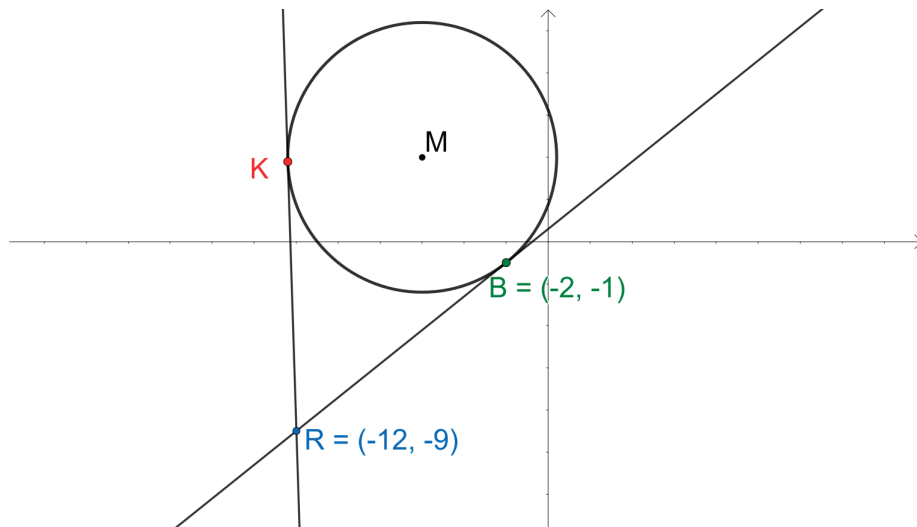
- b. What is the perimeter of $\triangle CMV$?

- c. Complete the table.

Line	$\overleftrightarrow{GW}$	$\overleftrightarrow{CM}$
Slope		
Equation in Point-Slope Form		
Equation in Slope-Intercept Form		

- d. Use the slope-intercept form of both lines to find point T .

3. Circle M is shown with tangents $\overline{RK}$ and $\overline{RB}$.



- What is the length of $\overline{RK}$?
- The center of the circle is located at $(-6, 4)$. Write the equation of circle M .
- What type of quadrilateral is $KMBR$? Justify your choice.
- What is the distance between point R and point M ?



13.2.4 Wrap-Up: Cool Down

- Write a note to a classmate who might have missed today's lesson, in your own words, about circles on the coordinate plane.